AI Agents

Tools, skills, memory, and autonomous action

What is an AI Agent?

An **AI agent** is an LLM that can take actions — not just respond.

- **Standard LLM:** Prompt → Response → Done
- **Agent:** Goal → Plan → Use Tools → Act → Report
- Ability to loop and retry until objective met
- Key difference: **Agency** (acting in the world)

The Agent Loop

Perceive
Receive goal & context

Plan
Decide steps & tools

Act
Call tool or action

Observe
See the result

Repeat
Until goal is met

Tools: Interaction Primitives

Search

Web browsing or database retrieval

Code

Writing and executing python/JS

API

Slack, Email, CRM connections

File System

Reading and writing local files

Browser

Navigating web pages directly

Math

Reliable arithmetic processing

Skills: Advanced Workflows

What are Skills?

Higher-level capabilities composed of multiple tools and logic.

Example: **Search + Summarize + Format** = "Topic Research" skill.

Implementation

Claude Code: /commit or /review-pr are built-in skills.

n8n: Sub-workflows used by agent nodes act as skills.

Tools are primitives. Skills are the workflows built on top.

Memory Types

Type	Description	Example
In-context	Text in current prompt window	Chat history
External	Database outside the model	Vector store, SQL
Episodic	Logs of past actions/outcomes	"Last time X failed"
Semantic	Stored facts and knowledge	User preferences

Planning Strategies

- **ReAct:** Synergizing Reason + Act (think, act, observe, repeat).
- **Plan-and-Execute:** Full strategy up front, then execution.
- **Tree of Thought:** Exploring multiple paths in parallel.
- **Reflection:** Self-critique of output before completion.

Most production agents use **ReAct** — it's simple, transparent, and robust.

Multi-Agent Systems

Splitting complex tasks across specialized agents:

Orchestrator: The manager who delegates.

Specialists: Researcher, Coder, Writer.

Critic: Quality assurance and reviews.

"Like a human team, each expert in their lane, coordinated by a lead."

Defining Success

Clear Goal

Avoid wandering behavior with precise instructions.

Right Tools

Capability is limited by the available toolset.

Memory

Persistent context across multiple steps.

Error Handling

Recovery logic for when tools fail.

Stop Conditions

Knowing when to stop and report back.

Architecture Summary

The Brain

LLM reasoning & planning engine.

Tools

Interface to the external world.

Memory

Context and history storage.

The Loop

Iterative execution cycle.

Agents in Action

From autonomous coding assistants to complex business automation, AI Agents are redefining workflow efficiency.

- **Claude Code:** Terminal-based coding agent.
- **n8n:** Visual agentic workflows.

Questions?

Exploring the future of autonomous systems.

```
> agent.status = "Task Completed"
```

Image Sources

Thumbnail for www.istockphoto.com

https://media.istockphoto.com/id/2152031133/photo/renovation-repair-concept-tools-3d-icon-on-the-abstract-human-hand-made-with-atom-array-and.jpg?s=612x612&w=0&k=20&c=RTsPJ8xi7b2U4kPyFhHkKnxAFnO5rEMUv8-ESDy_YFw=

Source: www.istockphoto.com

https://miro.medium.com/1*lufO0CP2e75CDY0VZWQl4w.png

Source: medium.com

Thumbnail for smythos.com

https://smythos.com/wp-content/uploads/2024/11/smart-industrial-control-system.jpeg.webp

Source: smythos.com

https://png.pngtree.com/background/20250316/original/pngtree-futuristic-digital-dashboard-with-real-time-data-analytics-and-visualizations-symbolizing-picture-image_16368178.jpg

Source: pngtree.com

https://motionarray.imgix.net/motion-array-3095463-pKX103aJae-high_0006.jpg?w=660&q=60&fit=max&auto=format

Source: motionarray.com